9 Two trolleys P and Q of masses $2m$ and m respectively move along the same straight line and collide. The total kinetic energy of the two trolleys is unchanged by the collision.

Before the collision, P moves to the right at 5.0 m s^{-1} and Q moves to the left at 4.0 m s^{-1}.

What is the velocity of P after the collision?

A 1.0 m s^{-1} to the left

B 1.0 m s^{-1} to the right

C 8.0 m s^{-1} to the left

D 8.0 m s^{-1} to the right